Bound to Forget?

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Introduction

- Conjunctive binding combines features into unique representations & relational binding links independent stimuli in memory[1,5]
- These involve different brain regions, are affected differently by ageing, represent distinct memory abilities
- Conjunctive binding may be more prone to interference[2]
- **(1) Are conjunctive and relational memory forgotten at different rates? (2) Is subjective memory, assessed via a questionnaire** (PRMQ[3]), **correlated with objective memory performance?**

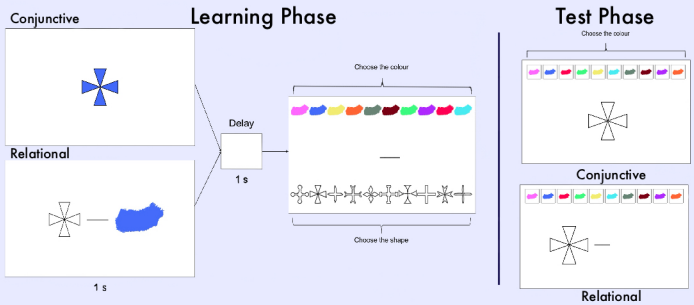
Hypotheses

1) Conjunctive bindings will be more rapidly forgotten over 1-day and 7-days than relational bindings.
2) Subjective memory (PRMQ) will not correlate with memory performance

Method

88 participants (x4 groups; n=20+/group)
Between-groups design: condition (conjunctive/relational) x delay (1-day/7-days)
All participants tested for immediate recall; half re-tested 1 day / 7 days later
→**Session 1) PRMQ,** binding task: **learning phase** & **test phase** (immediate recall) (in-person, 15 mins)
→**Session 2) test phase** (delayed recall) **& self-report debrief** (perceived difficulty & confidence) (remote, 5 mins)

PRMQ Example Item	Prospective/Retrospective	Short/Long-Term	Self/Environment Cued
Do you decide to do something in a few minutes time and then forget to do it?	Prospective	Short-Term	Self-Cued
Do you fail to recognise a place you have visited before?	Retrospective	Long-Term	Environmentally Cued



Results

1) **1-day delay:** relational group correctly recognised fewer pairs than the conjunctive group (Fig. a)
   7-day delay: no significant interaction between forgetting rates (Fig. a)
2) Subjective memory (PRMQ) not correlated with memory performance across immediate, 1-day or 7-day intervals

Overall, the relational group performed worse in immediate and delayed recall, felt less confident (Fig. c) and found it more difficult (Fig. d) than the conjunctive group

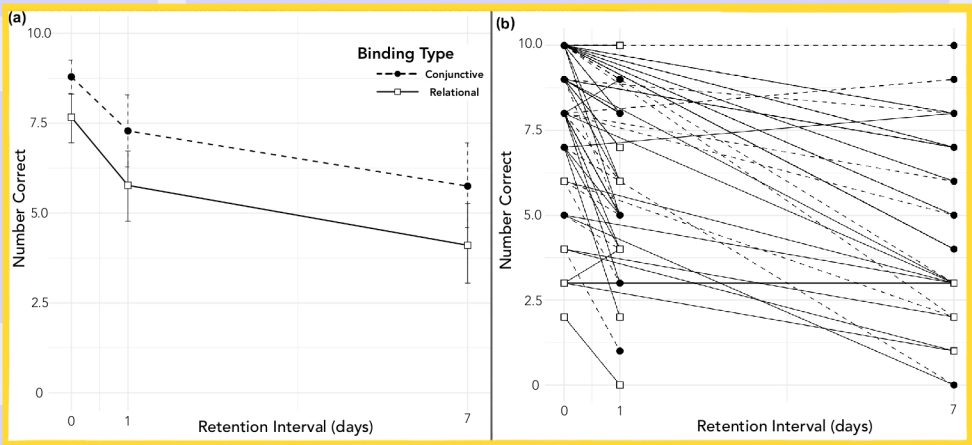


Figure (a) Mean & SE of Correct Responses at Each Binding Type and Retention Interval
(b) Total Correct Responses by Participant, Binding Type, and Retention Interval

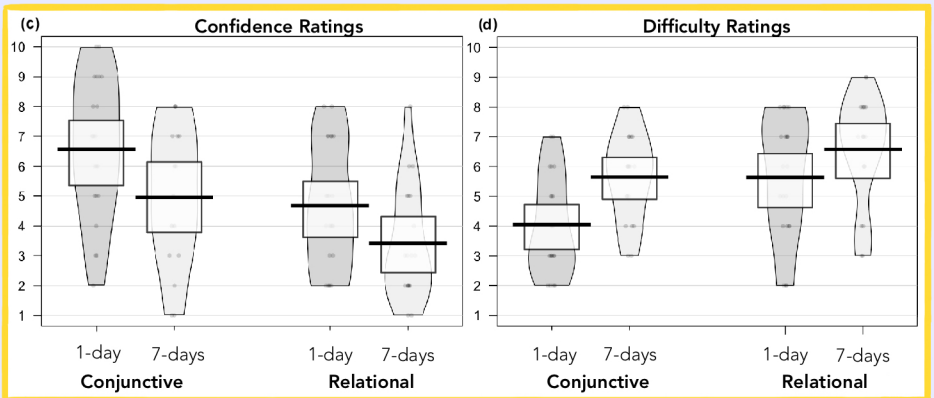


Figure (c) Distribution of confidence and (d) difficulty ratings (Mean & CI): wider sections indicate a higher density of data points

Conclusion

- Contrary to hyp 1: no significant difference in forgetting rates between conjunctive and relational binding
  - Relational group exhibited steeper forgetting after 1 day but after 7 days, the groups were equal
- Supports literature indicating no interaction between initial level of performance and forgetting rate[4]; despite qualitative differences, the same applies to binding types
- Hyp 2 supported: subjective memory did not correlate with memory ability

References

[1] Mayes, A., Montaldi, D., & Migo, E. (2007). Associative memory and the medial temporal lobes. Trends in Cognitive Sciences, 11(3), 126–135. Scopus. https://doi.org/10.1016/j.tics.2006.12.003
[2] Sadeh, T., Ozubko, J. D., Winocur, G., & Moscovitch, M. (2014). How we forget may depend on how we remember. Trends in Cognitive Sciences, 18(1), 26–36. https://doi.org/10.1016/j.tics.2013.10.008
[3] Smith, G., Della Sala, S., Logie, R. H., & Maylor, E. A. (2000). Prospective and retrospective memory in normal ageing and dementia: a questionnaire study. Memory (Hove, England), 8(5), 311–321. https://doi.org/10.1080/09658210050117735
[4] Rivera-Lares, K., Stamate, A., & Della Sala, S. (2021). The cognitive concept of forgetting. In G. Koob, & S. Della Sala (Eds.), Encyclopedia of Behavioural Neuroscience 2nd Edition: Reference Module in Neuroscience and Biobehavioral Psychology Elsevier Science. https://doi.org/10.1016/B978-0-12-819641-0.00155-9
[5] Wheeler, M. E., & Treisman, A. M. (2002). Binding in short-term visual memory. Journal of Experimental Psychology: General, 131(1), 48–64. https://doi.org/10.1037/0096-3445.131.1.48